PRODUCT USER MANUAL

For C3S Global L4 SST/IST analyses and ESA CCI SST analyses

SST_GLO_SST_L4_REP_OBSERVATIONS_010_024

Issue: 6.0

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	**2017/01/18**	All	Initial version (based on the ESA SST CCI Product User Guide)	Simon Good	
1.1	**2017/03/16**	Title page, section I	Added ESA and SST CCI logos and license details.	Simon Good	
2.0	**2019/04/12**		ESA SST CCI updated to version CDR2.0	Simon Good	
2.1	**2019/09/10**		C3S ICDR2.0 added	Simon Good	
2.2	**2020/09/10**	All	Document extension to 2019 and update document template	Simon Good	
2.3	**2021/01/15**	II.1, II.4	Temporal extensions	Simon Good	
3.0	**2022/06/16**	All	New template ; data updated to version 2.1.	Simon Good	Copernicus Marine Product management
4.0	**31/01/2025**	All	New template, Replacement product	Ioanna Karagali	Copernicus Marine Product management
5.0	**18/07/2025**	All	Reintroduce dataset ESACCI-GLO-SST-L4-REP-OBS-SST	Simon Good	Copernicus Marine Product management
6.0	**12/11/2025**	All	Updated version of dataset ESACCI-GLO-SST-L4-REP-OBS-SST	Simon Good	Copernicus Marine Product management

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the Product User Manual describing the Copernicus Climate Change Service (C3S) analysis (level 4) product over the global ocean, processed at the Danish Meteorological Institute, and the European Space Agency Sea Surface Temperature Climate Change Initiative (ESA SST CCI) global level 4 product, which was processed at the Met Office, UK. The data can be accessed through the Copernicus Marine Service in addition to the dissemination methods provided by C3S and ESA CCI. This manual covers how the dataset was created, its content, and which data services are available to access it through the Copernicus Marine Service.

The C3S reprocessed analysis product is a global, L4, Climate Data Record (CDR) combining Sea Surface Temperature (SST) and Sea-Ice Surface Temperature (IST) created using the DMIOI (Danish Meteorological Institute Optimal Interpolation) system (Høyer and She, 2007; Høyer et al., 2014; Nielsen-Englyst et al., 2023; Nielsen-Englyst et al., 2024) from re-processed L3U (A)ATSR, SLSTR and AVHRR SSTs while it also includes L2P Passive Microwave (PMW) SST from AMSR-E and AMSR-2. For the Sea Ice Surface Temperature (IST) it uses the Arctic and Antarctic Ice Surface Temperatures from AVHRR (AASTI) v2.1 and C3S L3 IST CDR v1.0 and ICDR v1.1. Its key features include its stability and its independence from in situ data. The CDR covers the period from 1982 to 2021 while the ICDR covers the period from January 2022 to December 2024. Data are on a global regular grid at 0.05° resolution. The product provides an estimate of the SST at 20 cm and IST skin.

The ESA SST CCI product is a satellite SST analysis created using the OSTIA (Operational SST and Sea Ice Analysis) system from re-processed (A)ATSR and AVHRR data (Embury et al., 2024). Its key features include its stability and its independence from in situ data. The ESA SST CCI data cover from 1st January 1980 to 31st December 2021. Data are on a global regular grid at 0.05° resolution. The product provides an estimate of the SST at 20 cm. The inputs to the system are SSTs at 10:30 am and pm local time which means that the analyses roughly correspond to the daily average SST.

Products are made available under a Creative Commons Attribution license (https://creativecommons.org/licenses/by/4.0/). This allows the sharing and adaptation of the data for any purpose as long as appropriate credit is given, a link to the license is provided and an indication of any changes made to the data is provided.

More information about this product can be found in:
The production system and the quality of the analyses are summarised in the Quality Information Document (QuID) - https://documentation.marine.copernicus.eu/QUID/CMEMS-SST-QUID-010-024.pdf.

The C3S documentation, including the licensing information, can be accessed at https://cds.climate.copernicus.eu/datasets/satellite-sea-surface-temperature?tab=download.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

C3S analyses :

Product Lines	SST_GLO_SST_L4_REP_OBSERVATIONS_010_024
Geographical coverage	Global : 180°W-180°E ; 90°S – 90°N
Variables	• Surface temperature • Sea surface temperature • Surface temperature uncertainty • Sea surface temperature uncertainty • Land Mask • Sea Ice Fraction • Sea Ice Fraction Flag
Product Type	Analysis
Available time series	01/01/1982 – 31/12/2024
Temporal resolution	Daily - Monthly means
Update Frequency	N/A
Target delivery time	N/A
Horizontal resolution	1/20 degree
Number of vertical levels	Surface only
Format	NetCDF4,Unidata Observation Dataset v1.0, GHRSST GDS v2.1

ESA SST CCI analyses :

Product Lines	SST_GLO_SST_L4_REP_OBSERVATIONS_010_024
Geographical coverage	Global : 180°W-180°E ; 90°S – 90°N
Variables	• Sea surface temperature • Sea surface temperature uncertainty

	• Mask • Sea Ice Fraction
Product Type	Analysis
Available time series	01/01/1980 – 31/12/2021
Temporal resolution	Daily
Update Frequency	N/A
Target delivery time	N/A
Horizontal resolution	1/20 degree
Number of vertical levels	Surface only
Format	netCDF-4

b) Production System Description
C3S analyses:

The algorithm documented here describes the steps followed to generate the L4 SST/IST product. The same algorithm is used for both the CDR and ICDR products. A schematic overview of the DMI Optimal Interpolation (DMIOI) system where all input data on surface temperatures over sea, sea ice and the marginal ice zone along with the input SIC fields and the land mask are combined into the final OI L4 product is shown below.

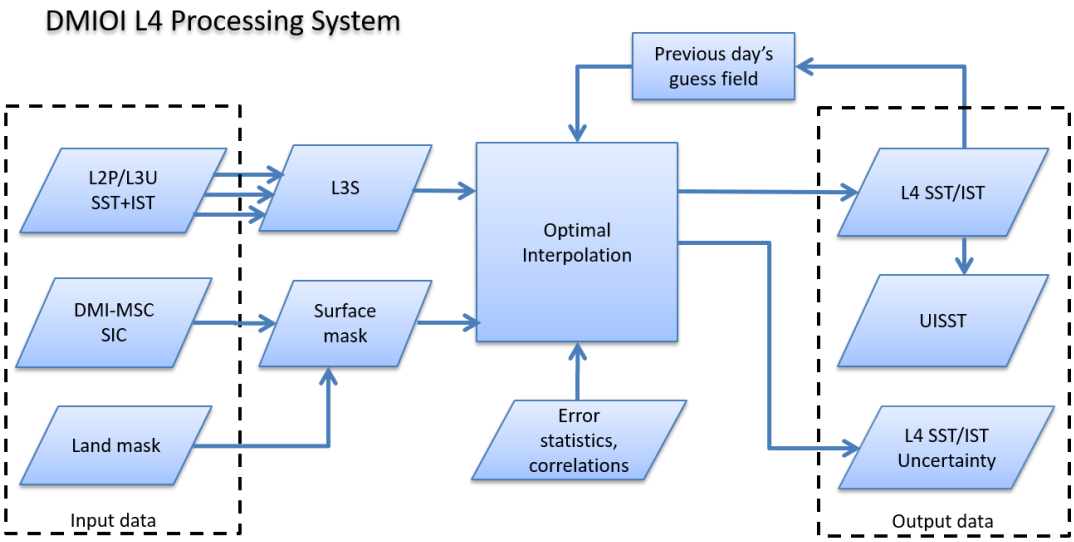


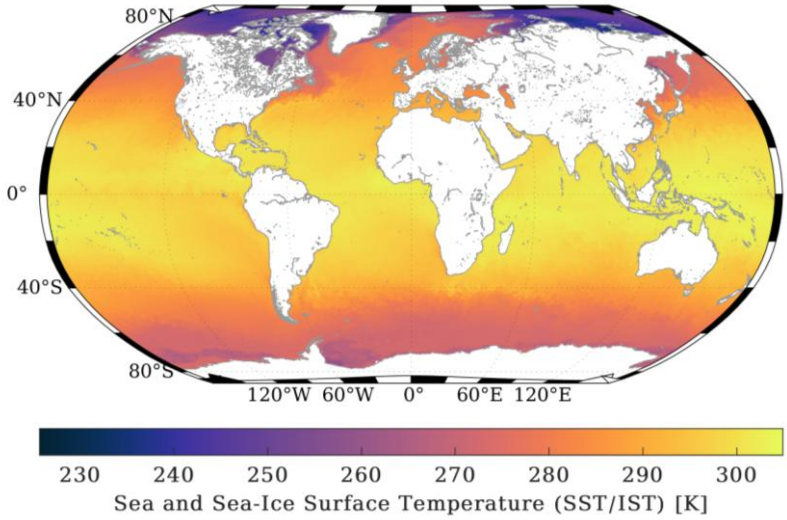
Figure 1: Schematic diagram of the DMIOI processing chain at DMI. Level 2 and Level 3 (L2/L3) SST and IST data, Sea Ice Concentration data (DMI-MSC-SIC) and land/sea data undergo a series of processes and finally an Optimal Interpolation methodology is applied to produce the Level 4 SST/IST data and its ancillary uncertainty dataset. An under-ice SST (UISST) is produced in the post-processing. Rectangles indicate processes, while slanted boxes indicate data products.

The final L4 analysis product is a merged and interpolated daily field with a 0.05° resolution in latitude and longitude, and covers both open ocean, sea ice and the Marginal Ice Zone (MIZ). The optimal interpolation method used to construct the merged and gap free SST/IST analysis is taken from the high latitude SST DMI processing scheme described in (Høyer and She (2007); Høyer et al. (2014); Nielsen-Englyst et al., 2023).

A high resolution land mask is used, which is a modified version of the Globolakes dataset (Carrea et al., 2015; GloboLakes: high-resolution global limnology dataset v1) obtained through the ESA SST_cci project (Globolakes-static_distance_to_land_Map-300m-P5Y-2005-ESACCI_WB-fv1.0_RES120_125km.nc) after personal communication with the Project Lead.

GLOBAL (180°W-180°E; 90°S – 90°N): 1/20°, regular grid, 7200 x 3600

This product is global with dedicated projection and spatial resolution. It is defined on a standard collocated grid at 1/20 degree (approx. 5 km). It covers the parameters Surface temperature (SST/IST), Surface temperature uncertainty, Sea Surface Temperature, SST uncertainty, ice concentration and mask variables which are provided on global, spatially complete (excluding lakes), regular grids.



Vertical grid	Surface Only
System Name	DMIOI
Model version	L4 SST/IST CDR/ICDR v1.0
Horizontal Resolution	1/20°
Vertical Resolution	1
Interpolation method	Optimal interpolation
Coupling	N/A
Sea Ice Model	N/A
Bathymetry	N/A
Initial conditions	N/A
Atmospheric forcings	N/A
Tidal constituents	N/A
Rivers	N/A
Data Assimilation scheme	N/A
Assimilated observations	SST input data representative of the 20cm depth temperature from the L3U AVHRR, (A)ATSR, SLSTR, AMSR-E, AMSR-2 v3.0 of ESA SST_cci CDR (1982-2021, Embury et al. 2024) and ICDR (funded by C3S for the 2022 extension and by UK Earth Observation Climate Information Service (EOCIS) and Marine and Climate Advisory Service (MCAS) from 2023 onwards). IST input data representative of sea-ice skin temperature were the L2P AASTI (Arctic and Antarctic Ice Surface Temperatures from Thermal Infra-red satellite) v2.1 CDR for

	1982-June 2019 and C3S IST ECV v1.1 ICDR from July 2019 onwards. A new sea ice concentration (SIC) dataset produced for the Copernicus Arctic Regional Reanalysis 2nd generation (CARRA2) Pan-Arctic extension (https://climate.copernicus.eu/copernicus-arctic-regional-reanalysis-service, referred to as DMI-MSC SIC) is used by the L4 analysis system for the northern hemisphere, while a rerun of the CARRA2 SIC processing has been produced for the southern hemisphere. The SIC dataset is a multi-source SIC composite based on SIC information from the EUMETSAT Ocean and Sea Ice Satellite Application Facility (OSI SAF) and ESA CCI Sea Ice products combined with different sources of ice charts. The following products are included in the SIC processing: 1. OSI SAF global SIC CDR v3, OSI-450a, used for the period 1982 to 1991. 2. The ESA CCI Sea ice CDR, SICCI-HR-SIC, used from 1992 to 2002 and in the gap in the AMSR satellite record, which occurs between the end of AMSR-E operations in October 2011 and the beginning of AMSR2 operations in May 2012. 3. OSI-458 based on AMSR satellite record and used for the period from 2002 to 2020 4. A special rerun of OSI-458 produced by METNO to produce consistent sea ice fields after 2020. 5. NIC ice charts used to identify land fast ice and to filter spurious sea ice along coasts 6. SHMI and FMI ice charts used for the Baltic region for the entire period 7. ESA SST_cci CDR v3 used to filter spurious sea ice

ESA SST CCI analyses :

The Operational Sea surface Temperature and Ice Analysis (OSTIA) reprocessing system has been run at the Met Office to produce the analyses. Figure 2 shows the different steps for the creation of the reprocessed products. Key information is provided in the table below.

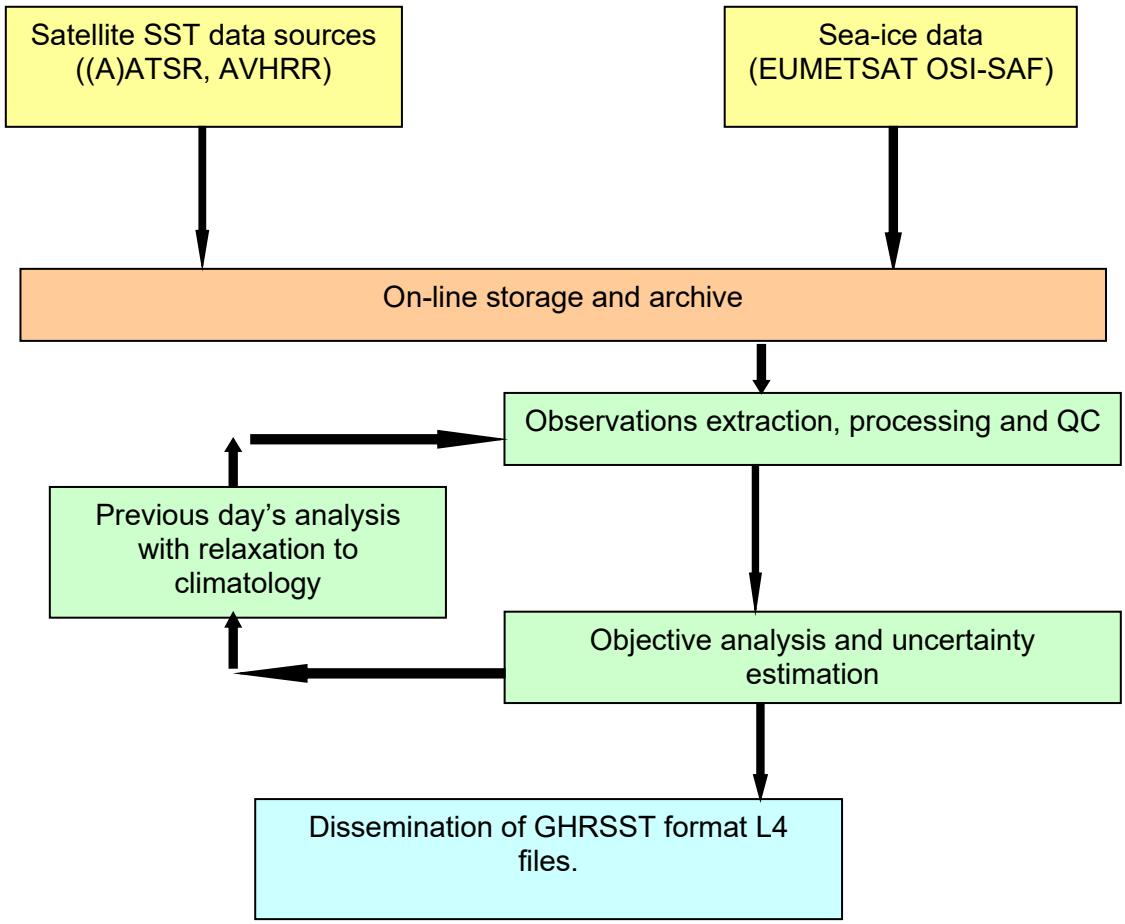
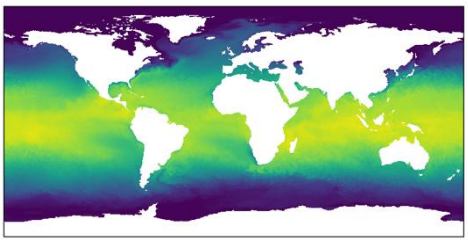


Figure 2. Schematic diagram of the reprocessing chain used for the ESA SST CCI processing.

Domain Resolution and grid Geographic coverage	GLOBAL (180°W-180°E ; 90°S – 90°N) with 1 time step per file 1/20˚ ; regular grid ; 7200 x 3600 x 1 SST, SST uncertainty, ice concentration and mask variables are provided on global, spatially complete (excluding lakes), regular grids. 
Version	CDR3.0
Assimilation scheme	NEMOVAR
Assimilated observations	SST input data were L3U AVHRR and (A)ATSR produced by the ESA SST CCI projects that are adjusted to 20 cm depth and 10.30 local time (Embury et al., 2024). Ice concentration data from the EUMETSAT OSI-SAF were used in the OSTIA reprocessing. Products used were:

	• OSI-450 - EUMETSAT Ocean and Sea Ice Satellite Application Facility. Global sea ice concentration climate data record 1979-2015 (v2.0, 2017), [Online]. Norwegian and Danish Meteorological Institutes. doi: 10.15770/EUM_SAF_OSI_0008. • OSI-430-b - EUMETSAT Ocean and Sea Ice Satellite Application Facility. Global sea ice concentration interim climate data record 2016 onwards (v2.0, 2019), [Online]. Norwegian and Danish Meteorological Institutes. Available from https://osisaf-hl.met.no/. Reference: Lavergne, T., Sørensen, A. M., Kern, S., Tonboe, R., Notz, D., Aaboe, S., Bell, L., Dybkjær, G., Eastwood, S., Gabarro, C., Heygster, G., Killie, M. A., Brandt Kreiner, M., Lavelle, J., Saldo, R., Sandven, S., and Pedersen, L. T.: *Version 2 of the EUMETSAT OSI SAF and ESA CCI sea-ice concentration climate data records*, The Cryosphere, 13, 49-78, doi:10.5194/tc-13-49-2019, 2019.
Climatology	ESA SST CCI climatology v2.2 (Embury et al., 2023)

c) Dataset details

C3S analyses :

Product Lines	SST_GLO_SST_L4_REP_OBSERVATIONS_010_024	
Geographical coverage	Global	
Datasets: C3S-GLO-SST-L4-REP-OBS-SST		
Product Type	Analysis	
Update frequency	Irregular	
Available time series	1982-2024	
Target delivery time	N/A	
Temporal resolution	Daily Monthly mean	

Dataset : C3S-GLO-SST-L4-REP-OBS-SST contains the 2D daily fields: analysed surface temperature, analysed surface temperature uncertainty , analysed sea surface temperature, analysed sea surface temperature uncertainty, sea ice fraction, sea ice fraction flag, mask,
analysed_st [K] analysed sea surface and_sea_ice_surface_temperature sea_water_sea_ice_temperature
analysed_sst [K] analysed sea surface temperature sea_water_temperature
analysed_st_uncertainty [K] estimated error standard deviation of analysed_st sea_water_sea_ice_temperature standard_error

analysed_sst_uncertainty [K] estimated error standard deviation of analysed_sst sea_water_temperature standard_error
sea_ice_fraction sea ice area fraction sea_ice_area_fraction
sea_ice_fraction_flag sea ice area fraction_flag sea_ice_area_fraction_flag
mask land sea ice lake bit mask

ESA SST CCI analyses:

Product Lines	SST_GLO_SST_L4_REP_OBSERVATIONS_010_024	
Geographical coverage	Global	
Datasets: ESACCI-GLO-SST-L4-REP-OBS-SST		
Product Type	Analysis	
Update frequency	Irregular	
Available time series	01/01/1980-31/12/2021	
Target delivery time	N/A	
Temporal resolution	Daily	

Dataset : ESACCI-GLO-SST-L4-REP-OBS-SST contains the 2D daily fields: analysed sea surface temperature, analysed sea surface temperature uncertainty, sea ice fraction, mask
analysed_sst [K] analysed sea surface temperature sea_water_temperature
analysed_sst_uncertainty [K] estimated error standard deviation of analysed_sst sea_water_temperature standard_error
sea_ice_fraction sea ice area fraction sea_ice_area_fraction
mask land sea ice lake bit mask

d) Additional information on some parameters

C3S analyses:

analysed _st [m]	Represents average daily sea surface temperature at 20 cm depth and sea-ice skin surface temperature.

analysed_sst [m]	Represents average daily temperature at 20 cm depth for the open water points. For sea-ice covered points, it represents the under-ice or sub-ice SST, estimated from an equation taking into account the climatological salinity,freezing point temperature of sea water and the sea ice concentration.
sea_ice_fraction_flag	Represents the flag of the sea ice concentration field, where 0 signifies the original SIC and 1 signifies any extrapolated parts.

ESA SST CCI analyses:

analysed_sst [K]	Represents average daily temperature at 20 cm depth. For sea-ice covered points, the temperature is relaxed to -1.8°C (271.35 K).

e) Processing information

C3S analysis product disseminating schedule :

Type of product (forecast/ hindcast)	Reference Time	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
Analysis	12 UTC	0 to 24	T-3months	Last weekday of month

f) Catalogue updating

C3S analyses:

The catalogue data from 1982 to 2021 (CDR) are not updated. The ICDR 2022-2024 is not updated. New ICDRs are foreseen every 3months depending on upstream data availability.

The product is foreseen to be updated regularly, at T-3 months intervals, by 1200 UTC of the last weekday of the month.

ESA SST CCI analyses:

This dataset is currently not updated. The dataset is foreseen to be updated regularly at 1 month delay to real time.

g) Time averaging

C3S analyses:

Dataset frequency	Time averaging
monthly	monthly means over the calendar month (first to last day of the month)
daily	The single sensor L3 fields produced by the DMIOI system prior to the optimal interpolation and production of the L4 analysis are calculated as the average of all available observations (meeting the quality level requirement) within the aggregation window. The aggregation window is 24 h for most of the record, except from the period 1982-1986, when limited input observations are available, and the window was extended to 60 h. The previous days analysis is used as a first guess field for the OI processing system.

ESA SST CCI analyses:

Dataset frequency	Time averaging
daily	The previous day's analysis and climatology are used to generate a first guess for the processing system. This is combined with observational input data from the analysis day and the two surrounding days. The uncertainty in the surrounding day's data is increased to reduce their influence on the analyses compared to the data from the analysis day.

h) Product use and impact

This product can be used to derive the Monitoring Indicators for the combined sea and sea-ice surface temperature annual anomalies and trends along with the time-series of monthly and annual anomalies. It is relevant for climate-type studies, where long-term, homogenised time-series of information is relevant and applicable.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
C3S-GLO-SST-L4-REP-OBS-SST	20210101120000-C3S2-L4_GHRSST-SSTdepth-ISTskin-DMIOI-GLOB_CDR-v02.0-fv01.0.nc	0.05
C3S-GLO-SST-L4-REP-OBS-SST	20220101120000-C3S2-L4_GHRSST-SSTdepth-ISTskin-DMIOI-GLOB_ICDR-v02.0-fv01.0.nc	0.05
ESACCI-GLO-SST-L4-REP-OBS-SST	20161231120000-ESACCI-L4_GHRSST-SSTdepth-OSTIA-GLOB_CDR3.0-v02.0-fv01.0.nc	0.016

4.2) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

5) REFERENCES

Quality Information Document
https://documentation.marine.copernicus.eu/QUID/CMEMS-SST-QUID-010-024.pdf

Høyer, J. L. and She, J. (2007) Optimal interpolation of sea surface temperature for the North Sea and Baltic Sea, J. Mar. Sys., Vol 65, 1-4, pp.

Høyer, J. L., Le Borgne, P., & Eastwood, S. (2014) A bias correction method for Arctic satellite sea surface temperature observations. Remote Sensing of Environment, 146, 201-213.

Nielsen-Englyst, P., Høyer, J. L., Kolbe, W. M., Dybkjar, G., Lavergne, T., Tonboe, R. T., Skarpalezos, S., Karagali, I. (2023) A combined sea and sea-ice surface temperature climate dataset of the Arctic, 1982–2021. Remote Sensing of Environment, 284, 113331, doi: https://doi.org/10.1016/j.rse.2022.113331.

Nielsen-Englyst, P., Høyer, J. L., Karagali, I., Kolbe, W. M., Tonboe, R. T., Pedersen, L. T. (2024) Impact of passive microwave observations on the estimation of Arctic sea surface temperatures. Remote Sensing of Environment, 301, 113949, doi: https://doi.org/10.1016/j.rse.2023.113949.

Embury, O., Merchant, C.J., Good, S.A., Rayner, N.A., Høyer, J.L., Atkinson, C., Block, T., Alerskans, E., Pearson, K.J., Worsfold, M., McCarroll, N., Donlon, C. Satellite-based time-series of sea-surface temperature since 1980 for climate applications. Scientific Data 11, 326 (2024). https://doi.org/10.1038/s41597-024-03147-w

Embury, O., Good, S. A. & Merchant, C. J. ESA Sea Surface Temperature Climate Change Initiative (SST_cci): Climatology Climate Data Record, version 2.2. NERC EDS Centre for Environmental Data Analysis
https://doi.org/10.5285/DC11996A68C446ABB342E917EFDAAC30 (2023)

Baltic Sea ice concentration, extent, and classification time series (SEAICE_BAL_PHY_L4_MY_011_019). E.U. Copernicus Marine Service Information (CMEMS). Marine Data Store (MDS). DOI: 10.48670/moi-00131 (Accessed on 30-Jan-2025)

The Recommended GHRSST-PP Data Specification GDS Version 2.1, https://doi.org/10.5281/zenodo.6984989

Carrea, L., Embury, O., and Merchant, C.J. (2015), GloboLakes: high-resolution global limnology dataset v1. Centre for Environmental Data Analysis, 21 July 2015. doi:10.5285/6be871bc-9572-4345-bb9a-2c42d9d85ceb.
https://dx.doi.org/10.5285/6be871bc-9572-4345-bb9a-2c42d9d85ceb

Lavergne, T., Sørensen, A. M., Kern, S., Tonboe, R., Notz, D., Aaboe, S., Bell, L., Dybkjær, G., Eastwood, S., Gabarro, C., Heygster, G., Killie, M. A., Brandt Kreiner, M., Lavelle, J., Saldo, R., Sandven, S., and Pedersen, L. T.: Version 2 of the EUMETSAT OSI SAF and ESA CCI sea-ice concentration climate data records, The Cryosphere, 13, 49-78, doi:10.5194/tc-13-49-2019, 2019.

6) ANNEX

a) For dataset C3S-GLO-SST-L4-REP-OBS-SST

```
netcdf 20210105120000-C3S2-L4_GHRSST-SSTdepth-ISTskin-DMIOI-GLOB_CDR1.0-v02.0-fv01.0 {

dimensions:

        lon = 7200 ;

        lat = 3600 ;

        time = 1 ;

variables:

        float lon(lon) ;

                lon:standard_name = "longitude" ;

                lon:long_name = "longitude" ;

                lon:units = "degrees_east" ;

                lon:axis = "X" ;

                lon:valid_range = -180.f, 180.f ;

                lon:comment = "geographical coordinates, WGS84 projection" ;

        float lat(lat) ;

                lat:standard_name = "latitude" ;

                lat:long_name = "latitude" ;

                lat:units = "degrees_north" ;

                lat:axis = "Y" ;

                lat:valid_range = -90.f, 90.f ;

                lat:comment = "geographical coordinates, WGS84 projection" ;

        int time(time) ;

                time:standard_name = "time" ;

                time:long_name = "reference time of sst field" ;

                time:axis = "T" ;
```

```
                time:comment = "Nominal time of Level 4 analysis" ;
                time:units = "seconds since 1981-01-01" ;
                time:calendar = "gregorian" ;
        short analysed_st(time, lat, lon) ;
                analysed_st:_FillValue = -32768s ;
                analysed_st:standard_name = "surface_temperature" ;
                analysed_st:long_name = "analysed sea and ice surface temperature" ;
                analysed_st:units = "kelvin" ;
                analysed_st:comment = "" ;
                analysed_st:source = "AASTI v2 SST/IST, ESA CCI SST and C3S SST L2P
products" ;
                analysed_st:valid_range = -6000s, 4500s ;
                analysed_st:add_offset = 273.15f ;
                analysed_st:scale_factor = 0.01f ;
        short analysis_error_st(time, lat, lon) ;
                analysis_error_st:_FillValue = -32768s ;
                analysis_error_st:long_name = "estimated error standard deviation of
analysed_st" ;
                analysis_error_st:units = "kelvin" ;
                analysis_error_st:comment = "" ;
                analysis_error_st:add_offset = 0.f ;
                analysis_error_st:scale_factor = 0.01f ;
        byte mask(time, lat, lon) ;
                mask:_FillValue = -128b ;
                mask:long_name = "land sea ice lake bit mask" ;
                mask:comment = "Mask can be used to further filter the data" ;
                mask:flag_meanings = "water land optional_lake_surface sea_ice
optional_river_surface" ;
                mask:source = "NAVOCEANO_landmask_v1.0, OSI-SAF, ESA CCI, CMEMS
                                Baltic and SMHI sea ice concentration" ;
```

mask:flag_masks = 1b, 2b, 4b, 8b, 16b ;

byte sea_ice_fraction(time, lat, lon) ;

sea_ice_fraction:_FillValue = -128b ;

sea_ice_fraction:standard_name = "sea_ice_area_fraction" ;

sea_ice_fraction:long_name = "sea ice area fraction" ;

sea_ice_fraction:units = "1" ;

sea_ice_fraction:comment = "" ;

sea_ice_fraction:source = "OSI-SAF, ESA CCI, NIC, CMEMS Baltic SMHI/FMI sea ice concentration" ;

sea_ice_fraction:valid_range = 0b, 100b ;

sea_ice_fraction:add_offset = 0.f ;

sea_ice_fraction:scale_factor = 0.01f ;

short analysed_sst(time, lat, lon) ;

analysed_sst:_FillValue = -32768s ;

analysed_sst:standard_name = "sea_surface_temperature" ;

analysed_sst:long_name = "analysed sea and under-ice surface temperature" ;

analysed_sst:units = "kelvin" ;

string analysed_sst:comment = "The under-ice SST is the freezing point temperature TS, computed as in Banzon et al. (2020), using Millero's formula (Fofonoff and Millard, 1983) and the World Ocean Atlas 2018 salinity climatology (Garcia et al., 2019) and further adjusted by a constant C and the sea ice concentration SIC." ;

analysed_sst:source = "AASTI v2 SST/IST, ESA CCI SST and C3S SST L2P products" ;

analysed_sst:valid_range = -6000s, 4500s ;

analysed_sst:add_offset = 273.15f ;

analysed_sst:scale_factor = 0.01f ;

byte sea_ice_fraction_flag(time, lat, lon) ;

sea_ice_fraction_flag:_FillValue = -128b ;

sea_ice_fraction_flag:long_name = "sea_ice_area_fraction_flag" ;

sea_ice_fraction_flag:comment = "The sea ice concentration flag is zero where the value is the original SIC value and one where the sea ice concentration field has been extrapolated." ;

sea_ice_fraction_flag:flag_meanings = "original, extrapolated" ;

sea_ice_fraction_flag:source = "OSI-SAF, ESA CCI, NIC, CMEMS Baltic SMHI/FMI sea ice concentration flag" ;

sea_ice_fraction_flag:flag_masks = 0b, 1b ;

sea_ice_fraction_flag:standard_name = "sea_ice_area_fraction_flag" ;

short analysis_error_sst(time, lat, lon) ;

analysis_error_sst:_FillValue = -32768s ;

analysis_error_sst:long_name = "estimated error standard deviation of analysed_sst" ;

analysis_error_sst:units = "kelvin" ;

analysis_error_sst:comment = "For the areas with under-ice SST computed, a fixed uncertainty has been assigned." ;

analysis_error_sst:standard_name = "estimated error standard deviation of analysed_sst" ;

analysis_error_sst:scale_factor = 0.01f ;

// global attributes:

:Conventions = "CF-1.4, Unidata Observation Dataset v1.0" ;

:title = "Global Sea and Ice Surface Temperature, L4, 5km daily reprocessed SST and IST" ;

:summary = "DMI Sea and Ice Surface Temperature analysis created using multisensor satellite surface temperature (ST) observations" ;

string :references = "Høyer, J. L. and She, J., Optimal interpolation of sea surface temperature for the North Sea and Baltic Sea, J. Mar. Sys., Vol 65, 1-4, pp. 176-189, 2007. Høyer, J. L., Le Borgne, P., & Eastwood, S., A bias correction method for Arctic satellite sea surface temperature observations, Remote Sensing of Environment, 146, 201-213, 2014. Nielsen-Englyst, P., Høyer, J. L., Kolbe, W. M., Dybkjær, G., Laverge, T., Tonboe, R. T., Skarpalezos, S., Karagali, I., A combined sea and sea-ice surface temperature climate dataset of the Arctic, 1982–2021, Remote Sensing of Environment, 284, 113331, 2023." ;

:institution = "Danish Meteorological Institute, DMI" ;

:history = "Version 1.0" ;

:license = "These data are available free of charge under the Copernicus Climate Change Service (C3S) data policy" ;

:id = "DMI-L4UHskin-GLOB-DMI_OI" ;

:naming_authority = "org.ghrsst" ;

:product_version = "Version 1.0" ;

:uuid = "d59e0252-bb82-45e1-ac85-04dbecc307c5" ;

:gds_version_id = "2.0" ;

:netcdf_version_id = "netcdf-4.4.1.1.6" ;

:date_created = "20240825T070856Z" ;

:file_quality_level = 3LL ;

:spatial_resolution = "0.05 degrees" ;

:time_coverage_start = "20210105T000000Z" ;

:time_coverage_end = "20210106T000000Z" ;

:geospatial_lon_min = -179.975 ;

:geospatial_lon_max = 179.975 ;

:geospatial_lat_min = -89.975 ;

:geospatial_lat_max = 89.975 ;

:source = "ESA SST CCI v3.0 and C3S v3.0 L3U, AASTIv2 IST L2P. OSI-SAF, ESA CCI, CMEMS FMI and SMHI Sea ice concentration" ;

:platform = "NOAA-6 to NOAA-19, ERS-1, ERS-2, ENVISAT, SENTINEL-3 A/B, MetOp A/B, AQUA, GCOM-W" ;

:platform_vocabulary = "CEOS mission table" ;

:instrument = "AVHRR, ATSR, ATSR-2, AATSR, SLSTR, AMSR-E, AMSR2" ;

:instrument_vocabulary = "CEOS instrument table" ;

:Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;

:metadata_link = "http://data.nodc.noaa.gov/NESDIS_DataCenters/metadata/index.html" ;

:keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;

:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;

:standard_name_vocabulary = "NetCDF Climate and forecast (CF) Metadata Convention" ;

:geospatial_lat_units = "degrees north" ;

:geospatial_lon_units = "degrees east" ;

:geospatial_lat_resolution = 0.05 ;

:geospatial_lon_resolution = 0.05 ;

:geospatial_bounds = "(-179.975, 179.975, -89.975, 89.975)" ;

:acknowledgment = "Please acknowledge the use of these data with the following statement: These data were provided by DMI and the Copernicus Climate Change Service (C3S) project" ;

:creator_name = "Danish Meteorological Institute - DMI Copenhagen" ;

:creator_email = "myocean_po@dmi.dk" ;

:creator_url = "http://ocean.dmi.dk/satellite/" ;

:project = "Copernicus Climate Change Service (C3S) project" ;

:publisher_name = "C3S and DMI" ;

:publisher_url = "https://climate.copernicus.eu/" ;

:publisher_email = "https://climate.copernicus.eu/help-and-support, myocean_po@dmi.dk" ;

:processing_level = "L4" ;

:cdm_data_type = "grid" ;

:comment = "IN NO EVENT SHALL DMI OR ITS REPRESENTATIVES BE LIABLE FOR ANY DAMAGES WHATSOEVER INCLUDING, WITHOUT LIMITATION, SPECIAL, INDIRECT, INCIDENTAL OR CONSEQUENTIAL DAMAGES OR DAMAGES FOR LOSS OF BUSINESS PROFITS OR SAVINGS, BUSINESS INTERRUPTION, LOSS OF BUSINESS INFORMATION OR OTHER PECUNIARY LOSS ARISING OUT OF THE USE OF OR THE BUSINESS INTERRUPTION, LOSS OF BUSINESS INFORMATION OR OTHER PECUNIARY LOSS ARISING OUT OF THE USE OF OR THE INABILITY TO USE THIS DMI PRODUCT, EVEN IF DMI HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. THIS LIMITATION SHALL APPLY TO CLAIMS OF PERSONAL INJURY TO THE EXTENT PERMITTED BY LAW. SOME COUNTRIES OR STATES DO NOT ALLOW THE EXCLUSION OR LIMITATION OF LIABILITY FOR CONSEQUENTIAL, SPECIAL, INDIRECT, INCIDENTAL DAMAGES AND, ACCORDINGLY, SOME PORTIONS OF THESE LIMITATIONS MAY NOT APPLY TO YOU. BY USING THIS PRODUCT, YOU HAVE ACCEPTED THAT THE ABOVE LIMITATIONS OR THE MAXIMUM LEGALLY APPLICABLE SUBSET OF THESE LIMITATIONS APPLY TO YOUR USE OF THIS PRODUCT. WARNING: Some applications are unable to properly handle signed byte values. If values are encountered > 127, please subtract 256 from this reported value" ;}

b) For dataset ESACCI-GLO-SST-L4-REP-OBS-SST

```
netcdf \20161231120000-ESACCI-L4_GHRSST-SSTdepth-OSTIA-GLOB_CDR3.0-v02.0-
fv01.0 {
dimensions:
      time = UNLIMITED ; // (1 currently)
      lat = 3600 ;
      lon = 7200 ;
      bnds = 2 ;
variables:
      int time(time) ;
            time:long_name = "reference time of sst field" ;
            time:standard_name = "time" ;
            time:axis = "T" ;
            time:calendar = "gregorian" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:comment = "" ;
            time:bounds = "time_bnds" ;
      int time_bnds(time, bnds) ;
            time_bnds:units = "seconds since 1981-01-01 00:00:00" ;
            time_bnds:long_name = "Time cell boundaries" ;
            time_bnds:comment = "Contains the start and end times for the
time period the data represent." ;
      float lat(lat) ;
            lat:standard_name = "latitude" ;
            lat:long_name = "Latitude" ;
            lat:units = "degrees_north" ;
            lat:valid_min = -90.f ;
            lat:valid_max = 90.f ;
            lat:actual_range = -89.975f, 89.975f ;
            lat:axis = "Y" ;
            lat:reference_datum = "geographical coordinates, WGS84 pro-
jection" ;
            lat:comment = "Latitude geographical coordinates, WGS84 pro-
jection" ;
            lat:bounds = "lat_bnds" ;
      float lat_bnds(lat, bnds) ;
            lat_bnds:units = "degrees_north" ;
            lat_bnds:long_name = "Latitude cell boundaries" ;
            lat_bnds:valid_min = -90.f ;
            lat_bnds:valid_max = 90.f ;
            lat_bnds:actual_range = -90.f, 90.f ;
            lat_bnds:comment = "Contains the northern and southern bound-
aries of the grid cells." ;
            lat_bnds:reference_datum = "geographical coordinates, WGS84
projection" ;
      float lon(lon) ;
            lon:standard_name = "longitude" ;
            lon:long_name = "Longitude" ;
            lon:units = "degrees_east" ;
            lon:valid_min = -180.f ;
            lon:valid_max = 180.f ;
            lon:actual_range = -179.975f, 179.975f ;
            lon:axis = "X" ;
            lon:reference_datum = "geographical coordinates, WGS84 pro-
jection" ;
```

```
            lon:comment = "Longitude geographical coordinates, WGS84 pro-
jection" ;
            lon:bounds = "lon_bnds" ;
      float lon_bnds(lon, bnds) ;
            lon_bnds:units = "degrees_east" ;
            lon_bnds:long_name = "Longitude cell boundaries" ;
            lon_bnds:valid_min = -180.f ;
            lon_bnds:valid_max = 180.f ;
            lon_bnds:actual_range = -180.f, 180.f ;
            lon_bnds:comment = "Contains the eastern and western bounda-
ries of the grid cells." ;
            lon_bnds:reference_datum = "geographical coordinates, WGS84
projection" ;
      short analysed_sst(time, lat, lon) ;
            analysed_sst:long_name = "analysed sea surface temperature" ;
            analysed_sst:standard_name = "sea_water_temperature" ;
            analysed_sst:units = "kelvin" ;
            analysed_sst:_FillValue = -32768s ;
            analysed_sst:add_offset = 273.15f ;
            analysed_sst:scale_factor = 0.01f ;
            analysed_sst:valid_min = -300s ;
            analysed_sst:valid_max = 4500s ;
            analysed_sst:actual_range = 271.15f, 305.18f ;
            analysed_sst:source = "ATSR<1,2>-ESACCI-L3U-v3.0, AATSR-
ESACCI-L3U-v3.0, SLSTR<A,B>-ESACCI-L3U-ICDR-v3.0
AVHRR<06,07,08,09,10,11,12,14,15,16,17,18,19>_G-ESACCI-L3U-v3.0,
AVHRRMT<A,B>-ESACCI-L3U-v3.0, AMSR<E,2>-ESACCI-L2P-v2.0" ;
            analysed_sst:depth = "20 cm" ;
            analysed_sst:ancillary_variables = "analysed_sst_uncertainty
mask" ;
      short analysed_sst_uncertainty(time, lat, lon) ;
            analysed_sst_uncertainty:long_name = "estimated error stand-
ard deviation of analysed_sst" ;
            analysed_sst_uncertainty:standard_name = "sea_water_tempera-
ture standard_error" ;
            analysed_sst_uncertainty:units = "kelvin" ;
            analysed_sst_uncertainty:_FillValue = -32768s ;
            analysed_sst_uncertainty:add_offset = 0.f ;
            analysed_sst_uncertainty:scale_factor = 0.01f ;
            analysed_sst_uncertainty:valid_min = 0s ;
            analysed_sst_uncertainty:valid_max = 32767s ;
            analysed_sst_uncertainty:actual_range = 0.11f, 2.62f ;
            analysed_sst_uncertainty:ancillary_variables = "mask" ;
      byte sea_ice_fraction(time, lat, lon) ;
            sea_ice_fraction:long_name = "sea ice area fraction" ;
            sea_ice_fraction:standard_name = "sea_ice_area_fraction" ;
            sea_ice_fraction:units = "1" ;
            sea_ice_fraction:_FillValue = -128b ;
            sea_ice_fraction:add_offset = 0.f ;
            sea_ice_fraction:scale_factor = 0.01f ;
            sea_ice_fraction:valid_min = 0b ;
            sea_ice_fraction:valid_max = 100b ;
            sea_ice_fraction:actual_range = 0.f, 1.f ;
            sea_ice_fraction:source = "EUMETSAT_OSI-SAF-ICE-OSI-450, EU-
METSAT_OSI-SAF-ICE-OSI-430-b" ;
            sea_ice_fraction:comment = " Sea ice area fraction" ;
            sea_ice_fraction:ancillary_variables = "mask" ;
```

```
      byte mask(time, lat, lon) ;
            mask:standard_name = "status_flag" ;
            mask:long_name = "sea/land/lake/ice field composite mask" ;
            mask:_FillValue = -128b ;
            mask:valid_min = 1b ;
            mask:valid_max = 31b ;
            mask:actual_range = 1b, 14b ;
            mask:flag_masks = 1b, 2b, 4b, 8b, 16b ;
            mask:flag_meanings = "water land optional_lake_surface
sea_ice optional_river_surface" ;
            mask:source = "Carrea, L.; Embury, O.; Merchant, C.J. (2015):
GloboLakes: high-resolution global limnology dataset v1. Centre for Envi-
ronmental Data Analysis, 21 July 2015. doi:10.5285/6be871bc-9572-4345-
bb9a-2c42d9d85ceb (described in: Carrea, L., Embury, O. and Merchant, C.
J. (2015) Datasets related to in-land water for limnology and remote
sensing applications: distance-to-land, distance-to-water, water-body
identifier and lake-centre co-ordinates. Geoscience Data Journal, 2 (2).
pp. 83-97. doi:10.1002/gdj3.32), Schaffer, J.; Timmermann, R. (2016):
Greenland and Antarctic ice sheet topography, cavity geometry, and global
bathymetry (RTopo-2), links to NetCDF files. PANGAEA, doi:10.1594/PAN-
GAEA.856844 (supplement to: Schaffer, J.; Timmermann, R.; Arndt, J.E.;
Kristensen, S.S.; Mayer, C.; Morlighem, M.; Steinhage, D. (2016): A
global, high-resolution data set of ice sheet topography, cavity geome-
try, and ocean bathymetry. Earth System Science Data, 8(2), 543-557,
doi:10.5194/essd-8-543-2016)" ;
            mask:comment = "b0: 1=grid cell is open sea water b1: 1=grid
cell is land b2: 1=grid cell is lake surface b3: 1=grid cell is sea ice
b4-b7: reserved for future grid mask data" ;

// global attributes:
            :Conventions = "CF-1.5, Unidata Observation Dataset v1.0" ;
            :title = "ESA SST CCI Analysis v3.0" ;
            :summary = "European Space Agency Sea Surface Temperature
Climate Change Initiative: Analysis product version 3.0" ;
            :references = "Embury, O. et al. Satellite-based time-series
of sea-surface temperature since 1980 for climate applications. Scien-
tific Data (2024). https://doi.org/10.1038/s41597-024-03147-w" ;
            :institution = "ESACCI" ;
            :history = "Created using OSTIA reanalysis system CDR3.0" ;
            :comment = "These data were produced at the Met Office as
part of the ESA SST CCI project. WARNING Some applications are unable to
properly handle signed byte values. If values are encountered > 127,
please subtract 256 from this reported value" ;
            :license = "Creative Commons Attribution 4.0 https://crea-
tivecommons.org/licenses/by/4.0/ \n",
                  "Users of these data should cite the dataset along with
the dataset paper: Embury, O. et al. Satellite-based time-series of sea-
surface temperature since 1980 for climate applications. Scientific Data
(2024). https://doi.org/10.1038/s41597-024-03147-w" ;
            :id = "OSTIA-ESACCI-L4-GLOB_CDR-v3.0" ;
            :naming_authority = "org.ghrsst" ;
            :product_version = "3.0.1" ;
            :uuid = "92fbdb08-855d-4eb1-bb74-ae8c7a184405" ;
            :tracking_id = "92fbdb08-855d-4eb1-bb74-ae8c7a184405" ;
            :gds_version_id = "2.0" ;
            :netcdf_version_id = "4.3.2" ;
            :date_created = "20220409T071011Z" ;
```

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            :time_coverage_start = "20161231T000000Z" ;
            :stop_time = "20170101T000000Z" ;
            :time_coverage_end = "20170101T000000Z" ;
            :time_coverage_duration = "P1D" ;
            :time_coverage_resolution = "P1D" ;
            :source = "ATSR<1,2>-ESACCI-L3U-v3.0, AATSR-ESACCI-L3U-v3.0,
SLSTR<A,B>-ESACCI-L3U-ICDR-v3.0
AVHRR<06,07,08,09,10,11,12,14,15,16,17,18,19>_G-ESACCI-L3U-v3.0,
AVHRRMT<A,B>-ESACCI-L3U-v3.0, AMSR<E,2>-ESACCI-L2P-v2.0, EUMETSAT_OSI-
SAF-ICE-OSI-450-v2.0, EUMETSAT_OSI-SAF-ICE-OSI-430-b" ;
            :platform = "ERS-<1,2>, Envisat, Sentinel-3<A,B>, NOAA-
<06,07,08,09,10,11,12,14,15,16,17,18,19>, MetOp<A,B>, Aqua, GCOM-W" ;
            :sensor = "ATSR, AATSR, SLSTR, AVHRR, AMSR" ;
            :Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
            :metadata_link =
"https://doi.org/10.5285/4a9654136a7148e39b7feb56f8bb02d2" ;
            :doi = "10.5285/4a9654136a7148e39b7feb56f8bb02d2" ;
            :keywords = "Oceans > Ocean Temperature > Sea Surface Temper-
ature" ;
            :keywords_vocabulary = "NASA Global Change Master Directory
(GCMD) Science Keywords" ;
            :standard_name_vocabulary = "NetCDF Climate and Forecast (CF)
Metadata Convention" ;
            :geospatial_lat_min = -90.f ;
            :geospatial_lat_max = 90.f ;
            :geospatial_lat_units = "degrees_north" ;
            :geospatial_lat_resolution = 0.05f ;
            :geospatial_lon_min = -180.f ;
            :geospatial_lon_max = 180.f ;
            :geospatial_lon_units = "degrees_east" ;
            :geospatial_lon_resolution = 0.05f ;
            :northernmost_latitude = 90.f ;
            :southernmost_latitude = -90.f ;
            :easternmost_longitude = 180.f ;
            :westernmost_longitude = -180.f ;
            :geospatial_vertical_min = -0.2f ;
            :geospatial_vertical_max = -0.2f ;
            :acknowledgment = "The European Space Agency (ESA) funded the
research and development of software to generate these data (grant refer-
ence ESA/AO/1-9322/18/I-NB), in addition to funding the production of the
Climate Data Record (CDR) for 1980 to 2021. The Copernicus Climate Change
Service (C3S) funded the development of the Interim-CDR (ICDR) extension
and production of ICDR during 2022. From 2023 onwards the production of
the ICDR is funded by the UK Natural Environment Research Council (NERC
grant reference number NE/X019071/1, Earth Observation Climate Infor-
mation Service) and the UK Marine and Climate Advisory Service (UKMCAS),
benefitting from the Earth Observation Investment Package of the Depart-
ment of Science, Innovation and Technology." ;
            :creator_name = "Met Office" ;
            :creator_url = "https://www.metoffice.gov.uk" ;
            :creator_email = "ml-ostia@metoffice.gov.uk" ;
            :creator_type = "institution" ;
            :creator_institution = "Met Office" ;
```

```
            :project = "Climate Change Initiative - European Space
Agency" ;
            :contributor_name = "JASMIN" ;
            :contributor_role = "This work used JASMIN, the UK\'s collab-
orative data analysis environment (https://jasmin.ac.uk)" ;
            :publisher_name = "NERC EDS Centre for Environmental Data
Analysis" ;
            :publisher_url = "https://www.ceda.ac.uk" ;
            :publisher_email = "support@ceda.ac.uk" ;
            :publisher_type = "institution" ;
            :processing_level = "L4" ;
            :cdm_data_type = "grid" ;
            :product_specification_version = "SST_CCI-PSD-UKMO-201-Issue-
2" ;
            :key_variables = "analysed_sst,sea_ice_fraction" ;
            :contact = "https://climate.esa.int/en/projects/sea-surface-
temperature" ;
}
```